Linux Basics

Introduction

Imagine you have a **computer without a graphical interface**. Instead of clicking on icons, you type commands to:

- **Create and manage files**
- **Run programs efficiently**
- **Control system processes**

This is how **Linux works**, and it powers **servers, cloud platforms, and even Android phones**!

Section 1: Learn

What is Linux?

Linux is an open-source operating system that allows users to interact with their computer using a **command-line interface (CLI). It is widely used in servers, cloud computing, and cybersecurity.**

Why Do We Need Linux?

Without Linux	**With Linux**
Expensive software licenses	**Free and open-source**
Limited customization	**Highly customizable**
Frequent crashes	**Stable and reliable**

Without Linux	With Linux
Vulnerable to viruses	Secure against most threats

Basic Linux Structure

A Linux system consists of:

- **Kernel – The core of the operating system that manages hardware.**
- **Shell – A command-line interface for interacting with Linux.**
- **File System – Organizes files in a hierarchical structure.**

How Does Linux Work?

1. **You type a command in the terminal.**
2. **The shell interprets the command and sends it to the kernel.**
3. **The kernel executes the command and returns the result.**

An Interesting Anecdote

In 1991, Linus Torvalds, a Finnish student, developed Linux as a free alternative to Unix. Today, Linux powers 90% of cloud servers, supercomputers, and Android devices.

Section 2: Practice

Step-by-Step Guide to Using Linux

1. Checking Your Linux Version

```
# Check the Linux distribution and version
lsb_release -a
```

```
# Check system details
uname -a
```

Why is this important?

- Helps identify which Linux version you are using.
- Useful for installing compatible software.

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2. Basic Linux Commands

Command	Purpose
pwd	Show current directory
ls	List files in a directory
cd directory_name	Change directory
mkdir folder_name	Create a new folder
rm filename	Delete a file
rmdir folder_name	Remove an empty folder
cp source destination	Copy files
mv old_name new_name	Rename/move files

Example Usage:

```
# Navigate to the home directory
cd ~

# Create a new folder
mkdir my_project

# List files inside the folder
ls my_project
```

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3. Managing Files and Directories

```
# Create a new file
touch myfile.txt

# Edit the file using nano editor
nano myfile.txt

# Display the contents of the file
cat myfile.txt
```

How does this help?

- Enables efficient file management.
- Helps in editing and viewing files easily.

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4. User Management Commands

Command	Purpose
whoami	Show current user
id	Display user ID and group info
sudo	Run commands as administrator
passwd	Change user password
adduser username	Add a new user
deluser username	Delete a user

Example Usage:

```
# Check current user
whoami


# Create a new user (requires sudo access)
sudo adduser student
```

Why is this useful?

- Helps in user authentication and security.
- Allows admin control over system users.

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5. Process and System Monitoring

```
# Show running processes
ps aux
```

```
# Check system memory usage
free -h

# Monitor system performance
top
```

How does this help?

- Allows monitoring of CPU, memory, and processes.
- Helps in troubleshooting system performance issues.

Real-World Example

A web hosting company uses Linux servers to run websites. Their system administrators use Linux commands to:

1. Monitor server performance and ensure uptime.
2. Secure websites against hacking attempts.
3. Install and configure software for different clients.

Without Linux:

- They would need expensive software licenses.
- Manual troubleshooting would increase downtime.

With Linux:

- Automated tasks save time.
- Servers run 24/7 without crashes.

Result? A cost-effective, stable, and secure web hosting service.

Section 3: Know More

Frequently Asked Questions

1. **Is Linux better than Windows?**
    - **For personal use: Windows is easier to use.**
    - **For servers and programming: Linux is more secure and powerful.**
2. **Can I use Linux on my laptop?**

    Yes! You can install Ubuntu, Fedora, or Mint as an alternative to Windows.
3. **What is the root user in Linux?**
    - **The root user is the administrator with full system access.**
    - **Use** sudo **to run commands as root safely.**
4. **How do I install software in Linux?**

```
sudo apt install software-name  # For Ubuntu/Debian
sudo yum install software-name  # For RHEL/CentOS
```

5. **Where can I practice Linux commands?**
    - **Use an online Linux terminal like LinuxCommand.org.**
    - **Install Ubuntu on VirtualBox and practice safely.**

Conclusion

Linux is a powerful, free, and secure operating system used by developers, companies, and governments worldwide.

Start with basic commands, explore file management, and soon you'll be using Linux like a pro!